Equivariant Flow Matching for Generative Cardiac Shape Modelling

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Abstract

[TODO: Write after results are in. Key claims to support: (1) SE(3)-equivariant mesh diffusion outperforms all four baselines on Chamfer distance and 1-NNA; (2) ejection fraction error under 5% absolute; (3) latent codes linearly separate the five ACDC pathology classes. Aim for 150–200 words.]

1 Introduction

[TODO: 3–4 paragraphs covering:]

- Clinical motivation: cardiac shape is a biomarker for disease (DCM, HCM, myocardial infarction).
- Gap: existing statistical shape models (PCA) are linear and voxel diffusion models discard surface topology.
- Our contribution: SE(3)-equivariant GNN combined with flow matching directly on cardiac meshes.
- Summary of results.

2 Background

2.1 Statistical Shape Models

Statistical shape models (SSMs) represent shape variation within a population by applying principal component analysis (PCA) to a set of registered point correspondences Cootes et al. [1995]. Given N shapes with V vertices each, we flatten each shape to a vector $\mathbf{s}_i \in \mathbb{R}^{3V}$ and compute the empirical covariance. New shapes are sampled as

$$\hat{\mathbf{s}} = \bar{\mathbf{s}} + \mathbf{P}\mathbf{b}, \quad \mathbf{b} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Lambda}), \tag{1}$$

where $\bar{\mathbf{s}}$ is the mean shape, $\mathbf{P}$ are the principal modes, and $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues. SSMs are computationally efficient but capture only linear variation, failing to represent non-linear deformation modes such as the asymmetric remodelling seen in hypertrophic cardiomyopathy.

2.2 Geometric Deep Learning and SE(3) Equivariance

A function f is *equivariant* under a symmetry group G if

$$f(g \cdot \mathbf{x}) = g \cdot f(\mathbf{x}) \quad \forall\, g \in G,\ \mathbf{x} \in \mathcal{X}. \tag{2}$$

For 3-D point sets, the relevant group is $\mathrm{SE}(3) = \mathrm{SO}(3) \ltimes \mathbb{R}^3$, the group of rotations and translations in three dimensions. A neural network operating on point positions that is equivariant to $\mathrm{SE}(3)$ will produce geometrically consistent outputs regardless of the coordinate frame in which the input is expressed.

We use the $E(n)$ *Equivariant Graph Neural Network* (EGNN) of Satorras et al. [2021] as our equivariant backbone. Each EGNN layer jointly updates scalar node features $\mathbf{h}_i$ and equivariant coordinates $\mathbf{x}_i$:

$$\mathbf{m}_{ij} = \phi_e\Big(\mathbf{h}_i,\, \mathbf{h}_j,\, \|\mathbf{x}_i - \mathbf{x}_j\|^2\Big)\,, \tag{3}$$

$$\mathbf{x}_i \leftarrow \mathbf{x}_i + \sum_{j \neq i} (\mathbf{x}_i - \mathbf{x}_j)\, \phi_x(\mathbf{m}_{ij}), \tag{4}$$

$$\mathbf{h}_i \leftarrow \phi_h\left(\mathbf{h}_i,\, \sum_j \mathbf{m}_{ij}\right)\,, \tag{5}$$

where ϕ_e, ϕ_x, and ϕ_h are multi-layer perceptrons. Because the coordinate update in (4) depends only on relative positions and the scalar ϕ_x, the overall mapping $(\mathbf{H}, \mathbf{X}) \mapsto (\mathbf{H}', \mathbf{X}')$ is equivariant to $\mathrm{SE}(3)$ by construction.

2.3 Diffusion Models and Flow Matching

Score-based diffusion models Song et al. [2021] learn to reverse a forward noising process, but typically require thousands of denoising steps at inference time. *Flow matching* Lipman et al. [2023] instead defines a deterministic probability path between a noise distribution p_0 and the data distribution p_1, parameterising the corresponding velocity field $v_\theta(\mathbf{x}_t, t)$ that transports samples from p_0 to p_1.

Under the *Optimal Transport* (OT) path Albergo and Vanden-Eijnden [2023], interpolating trajectories are straight lines:

$$\mathbf{x}_t = (1 - t)\, \mathbf{x}_0 + t\, \mathbf{x}_1, \quad t \sim \mathcal{U}[0, 1], \tag{6}$$

yielding the simple regression objective

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,\, \mathbf{x}_0 \sim p_0,\, \mathbf{x}_1 \sim p_1}\left[\|v_\theta(\mathbf{x}_t, t, \mathbf{z}) - (\mathbf{x}_1 - \mathbf{x}_0)\|^2\right], \tag{7}$$

where $\mathbf{z}$ is an optional conditioning variable. Samples are generated by integrating v_θ from $t = 0$ to $t = 1$ using a standard ODE solver (Euler or RK4). Compared to score-based methods, flow matching is faster to train (single forward pass per sample) and requires far fewer integration steps at inference.

3 Method

3.1 Data and Preprocessing

We use the *Automated Cardiac Diagnosis Challenge* (ACDC) dataset Bernard et al. [2018], which provides cine-MRI volumes with expert segmentation masks for 100 subjects spanning five pathology classes: normal (NOR), myocardial infarction (MINF), dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), and right ventricular abnormality (RV). Segmentations label the left ventricle (LV, label 3), right ventricle (RV cavity, label 1), and myocardium (Myo, label 2) at end-diastole (ED) and end-systole (ES).

Mesh extraction. For each structure at each frame, we apply the marching cubes algorithm Lorensen and Cline [1987] to the binary segmentation mask, using the physical voxel spacing as the step size to produce meshes in millimetre coordinates. We then apply ten iterations of Laplacian smoothing to reduce staircase artefacts, and decimate each mesh to approximately $2{,}500$ vertices via quadric edge collapse, targeting a balance between geometric fidelity and computational cost.

figures/ed_lv_single.pdf

(a) ED LV mesh after marching cubes, Laplacian smoothing, and quadric decimation ($\sim$2,500 vertices).

figures/ed_es_comparison.pdf

(b) Same subject at end-diastole (blue, larger) and end-systole (red, contracted). Volume difference gives the ejection fraction.

Figure 1: Representative LV surface meshes extracted from ACDC segmentation masks.

Template registration. To establish vertex correspondence across subjects we select a single reference subject and register all others to it. Registration proceeds in two stages: (i) a coarse Procrustes alignment using the Kabsch algorithm, and (ii) refinement with Iterative Closest Point (ICP). After alignment, each mesh is centred at the origin by subtracting its centroid, removing translation as a degree of freedom.

Graph construction. Each registered mesh is stored as a PyTorch Geometric `Data` object containing vertex positions $\mathbf{X} \in \mathbb{R}^{V\times 3}$, a k-nearest-neighbour edge index ($k = 8$) computed in the registered coordinate frame, the face connectivity for surface-area and volume calculations, and an integer pathology label $y \in \{0, \ldots, 4\}$.

3.2 Model Architecture

Our model consists of two learned components: a graph encoder that maps a cardiac mesh to a latent code, and a flow-matching velocity field conditioned on that code. Figure 2 provides an overview.

[TODO: Insert architecture diagram: mesh $\rightarrow$ encoder $\rightarrow$ z $\rightarrow$ velocity field $\rightarrow$ generated mesh]

Figure 2: Model architecture. The EGNN encoder compresses a cardiac mesh graph into a latent code $\mathbf{z}$. The EGNN velocity field, conditioned on $\mathbf{z}$ and time t, transports a noise cloud to a realistic cardiac mesh via OT flow matching.

EGNN Encoder. The encoder is a four-layer EGNN followed by mean-and-max global pooling, producing a Gaussian posterior $q_\phi(\mathbf{z} \mid \mathcal{G}) = \mathcal{N}(\boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2))$ with $\boldsymbol{\mu}, \log \boldsymbol{\sigma} \in \mathbb{R}^{64}$. Initial node features are set to a constant vector (no semantic node attributes are available), so all information must be inferred from the geometric structure of the graph.

Equivariant Velocity Field. The velocity field $v_\theta(\mathbf{x}_t, t, \mathbf{z})$ is a four-layer EGNN where each node's input feature is the concatenation of a sinusoidal time embedding and the conditioning code $\mathbf{z}$ projected to the hidden dimension. The network outputs a velocity vector in $\mathbb{R}^3$ for each vertex. Because position updates in EGNN depend only on relative coordinates, the velocity field inherits $\mathrm{SE}(3)$-equivariance: rotating the input mesh rotates the predicted velocities by the same rotation.

Training Objective. The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{CFM}}(\theta) + \beta \, D_{\text{KL}}(q_\phi(\mathbf{z} \mid \mathcal{G}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) \,, \tag{8}$$

where $\mathcal{L}_{\text{CFM}}$ is the OT flow matching loss from Equation (7) and $\beta = 10^{-3}$.

3.3 Baselines

We compare against four baselines, each designed to ablate a specific modelling choice.

Baseline 1: PCA Shape Model. Classical linear SSM fit to flattened vertex positions. Justifies the use of deep learning at all.

Baseline 2: Voxel Diffusion. OT flow matching applied to binary segmentation volumes on a 32^3 grid, using a 3-D U-Net velocity field. Justifies the mesh representation over voxel grids.

Baseline 3: PointNet++ VAE. PointNet++ encoder with a FoldingNet decoder trained as a VAE (no diffusion/flow). Justifies the distributional modelling component.

Baseline 4: Non-equivariant Flow Matching. Identical architecture to our main model, but EGNN layers replaced by standard MLP message passing; random rotation augmentation is applied during training to compensate. This is the *critical* ablation: it directly validates the geometric deep learning contribution.

4 Experiments

4.1 Setup

[TODO: Dataset split (70/15/15), hardware, training hyperparameters (epochs=200, lr=3e-4, batch=8, hidden=128, latent=64). Reference W&B project for full logs.]

4.2 Reconstruction Quality

[TODO: Table: Chamfer distance (mm) and Hausdorff-95 (mm) for all models. Expected finding: equivariant model $<$ MLP baseline $<$ PointNet++ $<$ PCA $\ll$ voxel.]

4.3 Clinical Validity

[TODO: Table: EF error mean $\pm$ std, % subjects with error $\leq 5\%$, myocardial volume conservation.]

Table 1: Reconstruction quality on the LV test set. Lower is better. $\pm$ denotes standard deviation across test subjects.

Model	Chamfer $\downarrow$	Hausdorff-95 $\downarrow$
PCA shape model (B1)	**[TODO:]**	**[TODO:]**
Voxel diffusion (B2)	**[TODO:]**	**[TODO:]**
PointNet++ VAE (B3)	**[TODO:]**	**[TODO:]**
MLP flow matching (B4)	**[TODO:]**	**[TODO:]**
Ours (EGNN + CFM)	**[TODO:]**	**[TODO:]**

Table 2: Clinical validity metrics.

Model	EF error (abs %) $\downarrow$	EF $\leq 5\%$ (%) $\uparrow$	Myo vol change (%) $\downarrow$
PCA (B1)	**[TODO:]**	**[TODO:]**	**[TODO:]**
MLP flow (B4)	**[TODO:]**	**[TODO:]**	**[TODO:]**
Ours	**[TODO:]**	**[TODO:]**	**[TODO:]**

4.4 Distribution Quality

[TODO: Table: 1-NNA (target ≈ 0.5) and pathology probe accuracy (%) for all generative models.]

Table 3: Distribution quality. 1-NNA closer to 0.5 is better.

Model	1-NNA $\rightarrow 0.5$	Pathology acc. (%) $\uparrow$
PCA (B1)	**[TODO:]**	**[TODO:]**
MLP flow (B4)	**[TODO:]**	**[TODO:]**
Ours	**[TODO:]**	**[TODO:]**

4.5 Ablation: Equivariance Under Rotation

[TODO: Show performance of main model vs. B4 as a function of training set size (e.g., 10%, 25%, 50%, 100% of training data). Expected finding: equivariant model degrades more gracefully in low-data regime.]

5 Discussion

[TODO: 3–4 paragraphs covering:

- **What the results say about the value of SE(3) equivariance.**
- **Limitations: single-structure, ED/ES only (not full cycle), Procrustes registration assumes rigid alignment.**
- **Future work: multi-structure joint model, 4D temporal modelling, learned registration.**

]

6 Conclusion

[TODO: 4–6 sentences. Restate problem, method, and three key findings.]

References

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